

Fake News Classification using TF-IDF and Linguistic Feature Analysis

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Qualitative Analysis

Background / Motivation

The rapid rise of digital media platforms has significantly increased the spread of fake and misleading news. These articles are often designed to attract attention using sensational headlines, emotional language, and exaggerated claims, making it difficult for users to distinguish between reliable and unreliable information. Traditional keyword-based filtering techniques are no longer effective because fake news creators continuously modify vocabulary to bypass such systems. Therefore, it is essential to develop intelligent models that go beyond simple word matching and instead analyze the writing style, tone, and structure of the text. This project addresses this challenge by building a fake news classifier that incorporates both lexical features and linguistic characteristics to detect subtle stylistic patterns associated with fake news, thereby improving reliability and trust in digital information systems.

Objectives

- To develop a machine learning model capable of classifying news articles as real or fake
- To extract and analyze linguistic features such as superlatives, pronouns, and adjectives
- To study syntactic complexity using sentence length and tree depth analysis
- To compare the performance of TF-IDF-only and combined feature-based models
- To interpret model behavior using visualization techniques and error analysis

Methods

This project utilizes the ISOT Fake News Dataset, which contains a collection of real and fake news articles. A balanced subset of 2000 articles (1000 real and 1000 fake) was selected to ensure fair and unbiased model training. The dataset was preprocessed using several natural language processing techniques to prepare it for feature extraction and classification.

The preprocessing phase included tokenization of text into sentences and words, followed by lemmatization to reduce words to their base forms. Unlike traditional approaches, stopwords such as pronouns were selectively retained, as they provide important stylistic cues. For example, fake news articles often use first-person pronouns to create a persuasive or emotional tone.

Feature engineering was performed using Part-of-Speech (POS) tagging to extract linguistic features. These included superlative ratio (to detect exaggeration), proper noun ratio (to measure specificity), pronoun ratio (to identify persuasive tone), and adjective count (to quantify emotional intensity). Additionally, syntactic features such as average sentence length and tree depth were computed to analyze the grammatical complexity of the text.

For classification, TF-IDF vectorization was used to convert textual data into numerical features. Logistic Regression was selected as the classification model due to its efficiency and effectiveness with high-dimensional sparse data. Two models were trained: one using only TF-IDF features and another combining TF-IDF with linguistic features. Model performance was evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Visualization techniques such as TF-IDF feature plots and PCA-based dimensionality reduction were also employed to interpret the results.

Results

The experimental results demonstrate that the TF-IDF-only model achieved an accuracy of approximately **96.00%**, while the combined model incorporating both TF-IDF and linguistic features achieved an accuracy of approximately **95.5%**. Both models showed strong and balanced performance across precision, recall, and F1-score, indicating that the classifier is effective in distinguishing between real and fake news articles.

The confusion matrix revealed a low number of misclassifications, with only a small proportion of false positives and false negatives. This indicates that the model performs consistently well across both classes. The classification report further confirmed that both classes were predicted with nearly equal precision and recall, suggesting no significant bias.

Visualization of TF-IDF features highlighted the most important words contributing to the classification process. After removing stopwords, the most relevant terms included

content-specific and context-driven words, which play a significant role in differentiating between fake and real news.

The PCA visualization of TF-IDF features showed partial clustering between the two classes. While some separation between fake and real news articles was observed, there was also considerable overlap. This is expected due to the reduction of high-dimensional data into two dimensions, which leads to loss of information.

Error analysis revealed that certain real news articles were incorrectly classified as fake due to their use of emotionally expressive language and shorter sentence structures. These characteristics resemble patterns commonly found in fake news, highlighting a limitation of style-based classification.

Conclusion

In conclusion, the project successfully demonstrates that TF-IDF-based features are highly effective for fake news classification, achieving high accuracy with a simple Logistic Regression model. While the addition of linguistic and syntactic features did not significantly improve performance, they provide valuable insights into writing style and enhance interpretability. The results highlight that lexical patterns are strong indicators in this dataset, while stylistic features may vary across articles. Future improvements could include the use of advanced models such as deep learning and larger datasets to capture more complex patterns and improve generalization.

Quantitative Analysis

2.1 Dataset and Preprocessing

The dataset used in this project is the ISOT Fake News Dataset, consisting of real and fake news articles. A balanced subset of 2000 samples (1000 real and 1000 fake) was selected to ensure unbiased model training. The dataset is textual in nature and written in English.

The preprocessing pipeline involved tokenization of text into sentences and words, followed by stopword filtering while retaining pronouns, and lemmatization to normalize word forms. This approach preserves stylistic features such as pronoun usage and emotional expressions, which are important indicators of fake news.

Table 2.1: Dataset Summary

Feature	Value
Dataset Name	ISOT Fake News Dataset
Total Samples	2000
Real News	1000
Fake News	1000
Data Type	Text
Language	English

2.2 Model Design / Algorithm Implementation

The classification model used in this project is Logistic Regression, which is well-suited for handling high-dimensional sparse data such as TF-IDF vectors. The model takes input features derived from both lexical and linguistic representations of the text.

TF-IDF vectorization was used with a maximum of 5000 features and English stopword removal. Linguistic features such as superlative ratio, proper noun ratio, pronoun ratio, adjective count, and average sentence length were incorporated to capture stylistic characteristics.

Table 2.2: Model Configuration

Component	Description
Model	Logistic Regression
Input Features	TF-IDF + Linguistic Features
TF-IDF Size	5000
Linguistic Features	5
Vectorizer	TF-IDF

Table 2.3: Hyperparameters

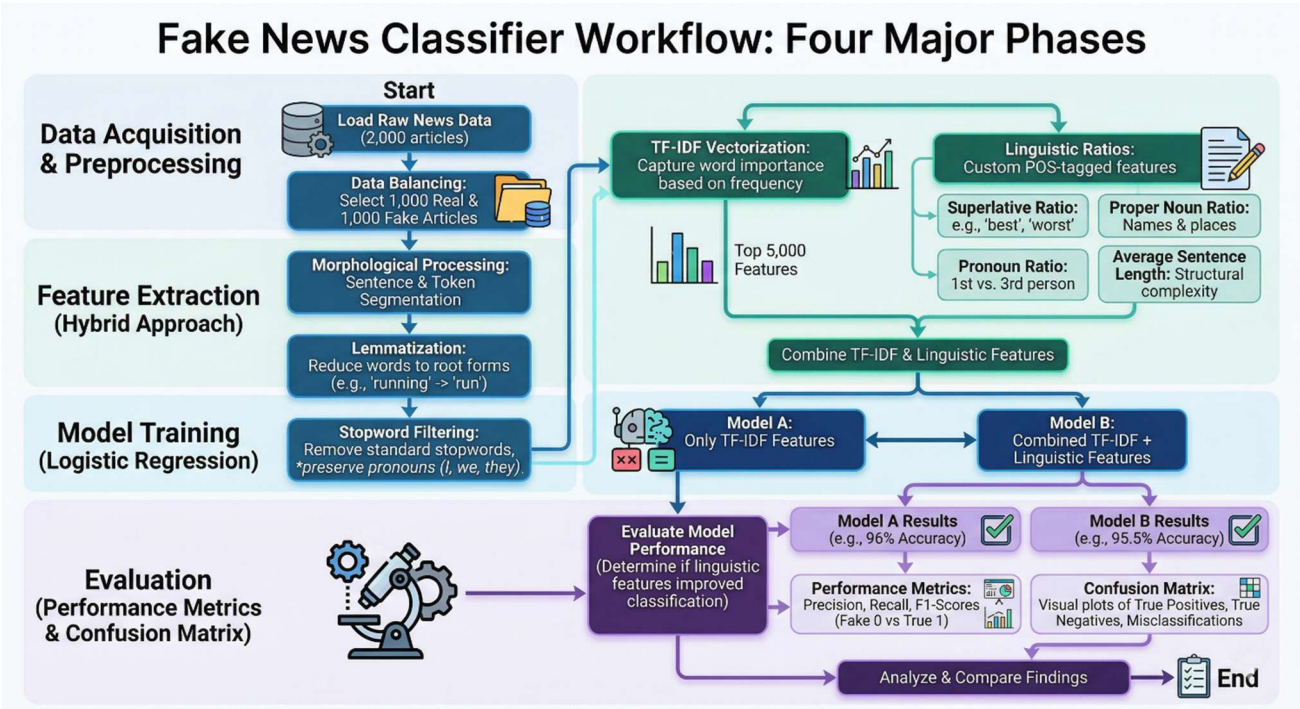
Parameter	Value
max_iter	200
Train-Test Split	80:20
Stopword Removal	Enabled
Feature Combination	TF-IDF + Linguistic

2.3 Training and Evaluation Setup

The dataset was split into training and testing sets using an 80:20 ratio. The model was trained on the training data and evaluated on unseen test data.

Performance was measured using accuracy, precision, recall, and F1-score. The confusion matrix was used to evaluate classification outcomes in terms of true positives, false positives, true negatives, and false negatives.

Figure 2.1: Overall Workflow of Fake News Classification System



2.4 Experimental Results

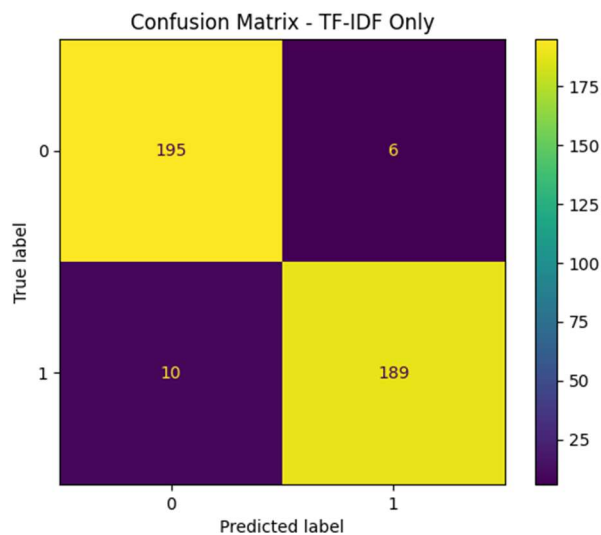
The performance of the models was evaluated using classification metrics and visual analysis.

Table 2.4: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
TF-IDF Only	0.960	0.969231	0.949749	0.959391
TF-IDF + Linguistic	0.955	0.944724	0.944724	0.954315

The TF-IDF-only model achieved slightly higher accuracy compared to the combined model. This indicates that lexical features alone are highly effective for this dataset.

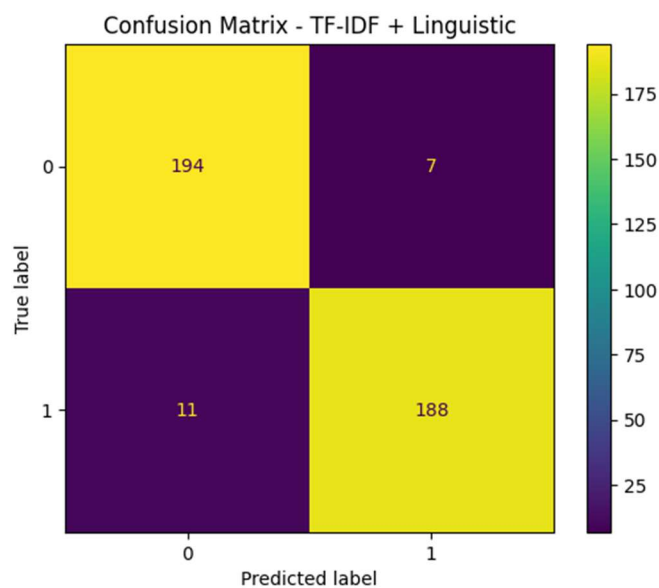
Figure 2.2: Confusion Matrix – TF-IDF Only Model



The confusion matrix for the TF-IDF-only model shows strong classification performance with a high number of correct predictions. Out of the total test samples, 195 fake news articles were correctly classified (True Negatives) and 189 real news articles were correctly identified (True Positives). The model made very few errors, with only 6 false positives (fake predicted as real) and 10 false negatives (real predicted as fake).

This indicates that the TF-IDF-only model is highly effective in distinguishing between fake and real news, achieving high accuracy with minimal misclassification.

Figure 2.3: Confusion Matrix – TF-IDF + Linguistic Model



The confusion matrix for the TF-IDF-only model shows strong classification performance with a high number of correct predictions. Out of the total test samples, 195 fake news articles were correctly classified (True Negatives) and 189 real news articles were correctly identified (True Positives). The model made very few errors, with only 6 false positives (fake predicted as real) and 10 false negatives (real predicted as fake).

This indicates that the TF-IDF-only model is highly effective in distinguishing between fake and real news, achieving high accuracy with minimal misclassification.

Overall, both models perform well, but the TF-IDF-only model shows slightly better accuracy and fewer misclassifications compared to the combined model.

2.4.1 Linguistic Profile Analysis

To analyze stylistic differences between fake and real news, linguistic features such as adjective count and sentence length were evaluated.

Table 2.5: Linguistic Feature Comparison

News Type	Average Adjective Count	Average Sentence Length
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Fake News	48.832	33.63
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Real News	47.336	31.25
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Interpretation:

Fake news articles exhibit a slightly higher average adjective count, indicating the use of more descriptive and emotionally expressive language. Additionally, fake news tends to have longer sentences on average, suggesting a tendency toward elaboration or exaggeration. However, the differences are relatively small, indicating that these features alone may not strongly distinguish between the two classes.

2.4.2 Syntax Analysis (Tree Depth)

To analyze grammatical complexity, tree depth was computed using dependency parsing. Tree depth represents the number of hierarchical levels in the syntactic structure of a sentence.

Table 2.6: Average Tree Depth Comparison

News Type	Average Tree Depth
Fake News	5.90
Real News	6.76

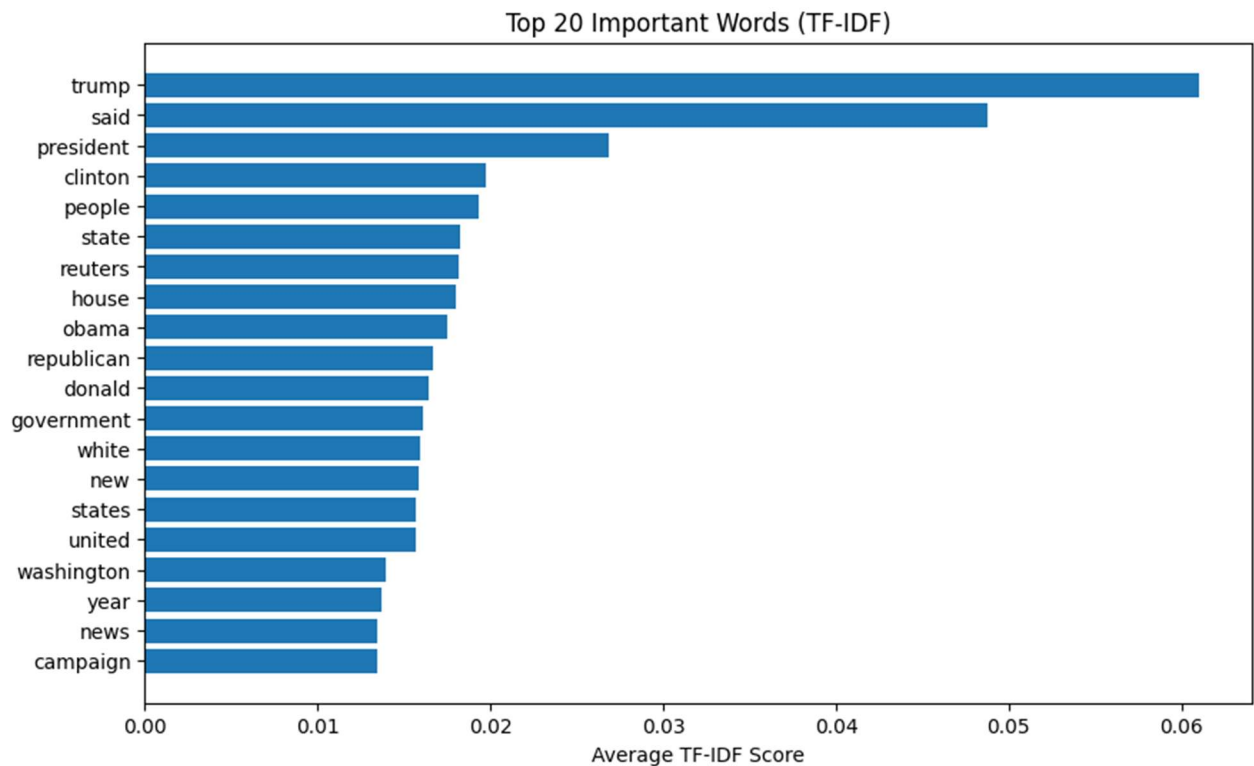
Interpretation:

The results show that real news articles have a higher average tree depth compared to fake news. This indicates that real news tends to use more complex sentence structures with embedded clauses, while fake news relies on simpler constructions. However, the difference is moderate, suggesting that syntactic complexity is not the dominant factor in classification.

2.5 Visual Results

Visualization techniques were used to interpret model behavior and feature distribution.

Figure 2.4: Top TF-IDF Feature Importance Plot

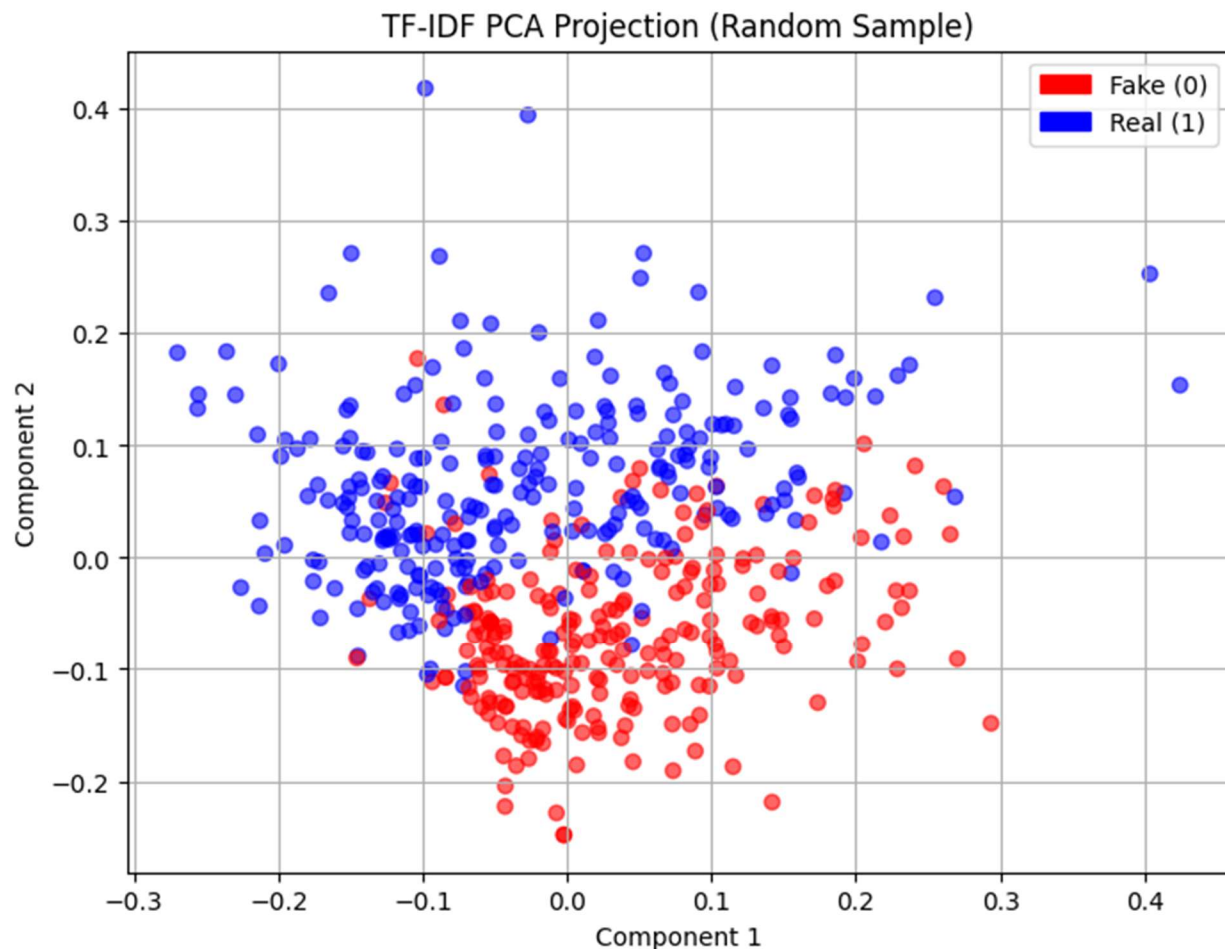


The TF-IDF feature importance plot highlights the top 20 most significant words based on their average TF-IDF scores across the dataset. These words represent the most informative and discriminative features used by the classification model.

The presence of words such as *“trump”*, *“said”*, *“president”*, *“clinton”*, and *“government”* indicates that the dataset is heavily influenced by political and news-related content. These terms are highly relevant in distinguishing between fake and real news articles, as they frequently appear in specific contexts within the dataset.

The relatively higher TF-IDF scores of these words suggest that they occur frequently in certain documents but are not uniformly distributed across all documents, making them useful for classification. This visualization confirms that lexical features play a dominant role in the model’s decision-making process.

Figure 2.5: PCA Visualization of TF-IDF Features



The PCA (Principal Component Analysis) visualization presents a two-dimensional projection of the high-dimensional TF-IDF feature space. Each point represents a news article, with red points corresponding to fake news and blue points representing real news.

The plot shows partial clustering of the two classes, where real news articles are more concentrated in the upper region, while fake news articles are more densely distributed in the lower region. However, there is noticeable overlap between the two classes, particularly around the central region of the plot.

This overlap indicates that the separation between fake and real news is not perfectly linear when reduced to two dimensions. This is expected because PCA reduces thousands of features into just two principal components, resulting in some loss of

information. Despite this, the observable clustering suggests that TF-IDF features capture meaningful patterns that help distinguish between the two classes in the original high-dimensional space.

These visualizations collectively demonstrate that while TF-IDF effectively captures important lexical patterns, dimensionality reduction techniques such as PCA may not fully preserve class separability, highlighting the importance of high-dimensional feature spaces in text classification tasks.

Figure 2.6: Misclassified News Article Example

TEXT:

“SOFIA (Reuters) - Couples lining up to face each other are given 30 seconds to chat before moving on - in a version of speed dating that, rather than sparking romance, aims to promote understanding and integration for new immigrants. At a social center in the Bulgarian capital Sofia, the locals stand on one side, the immigrants - many refugees from Syria, Afghanistan and other war-torn countries - face them. The quick-fire conversations - in which participants have just seconds to share their dreams and stories of love, pain and family - are aimed at starting a dialogue between two communities that in normal daily life rarely speak to each other on an equal footing. It s so good to have an event like this, especially for refugees, said Rodi Naamo, a 26-year-old Syrian Kurd who arrived in Bulgaria in 2015, when hundreds of thousands of migrants passed through the Balkans en route to Germany and northern Europe. Bulgarians also appreciated the speedy encounters, as well as the Middle Eastern food and traditional Afghan dances afterwards. I really enjoyed it, it helped me overcome prejudices, said Milka Bocheva, a Bulgarian who runs an online business. They (the refugees) are just people like us, we gave such similar answers. Bulgaria stands out in the region for its significant Muslim minority, some 12 percent among 7.1 million mainly Orthodox Christians - a legacy of its history as part of the Ottoman Empire. Despite that, large parts of the population are concerned about the influx of asylum-seekers and migrants. Naamo, who returned to Bulgaria after a short spell in Germany, to work in a call center, believes opinions are starting to change. I decided to stay here, this is my country. I ll bring my family here, he said.”

```
misclassified_real = results[
    (results["true"] == 1) &
    (results["pred"] == 0)
]

misclassified_real.head(1)
```

	text	true	pred
10	SOFIA (Reuters) - Couples lining up to face ea...	1	0

The misclassification analysis shows that some real articles were incorrectly classified as fake due to the presence of emotionally expressive language and stylistic similarities with fake news.

Analysis and Discussions:

- The results of this study demonstrate that TF-IDF-based lexical features are highly effective for fake news classification. The TF-IDF-only model achieved the highest accuracy (96.0%), outperforming the combined model that included linguistic features. This suggests that word-level patterns and term distributions are strong indicators of fake and real news in the given dataset. The confusion matrices further confirm this, showing very low false positives and false negatives, indicating robust and balanced classification performance.
- The addition of linguistic features such as adjective count, pronoun ratio, and sentence length did not significantly improve performance. While these features were expected to capture stylistic differences, their impact was limited due to their low dimensionality compared to the high-dimensional TF-IDF vectors. However, linguistic analysis revealed meaningful trends: fake news articles tend to use slightly more adjectives and longer sentences, indicating a tendency toward emotional and descriptive writing. Similarly, syntax analysis showed that real news articles have higher average tree depth, suggesting more complex grammatical structures.
- The PCA visualization showed partial clustering between fake and real news, but also significant overlap. This aligns with expectations, as dimensionality reduction leads to information loss, making perfect separation difficult in lower dimensions.
- Compared to typical benchmarks in text classification, where TF-IDF combined with linear models often performs strongly, the results are consistent with existing literature. Overall, the model demonstrates high accuracy and reliability, but also highlights that stylistic and syntactic features alone may not significantly enhance performance beyond lexical features in this dataset.